LESSON PLAN

ENGINEERING METROLOGY (10)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	10
Total No. of Lesson Plans	11	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Surface roughness, meaning of real surface, geometrical surface, effective surface, and various irregularities
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Surface roughness, meaning of real surface, geometrical surface, effective surface, and various irregularities.
5. Specific Objectives	10.1.1.1 Trainee will be able to describe details about Engineering Metrology.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of surface finish standards chart; Instructor supervision on terminology definitions
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary purpose of Engineering Metrology?
(b) Can you name one tool used for measuring surface finish?
(c) Why is surface finish critical in aircraft engine components?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of Real Surface • Definition of Geometrical Surface • Definition of Effective Surface • Distinction between Real and Geometrical • Role of Measurement Instruments	L & Ds	PPT, IB	15 min
(b)	• Classification of Surface Irregularities • Long Wavelength Irregularities • Short Wavelength Irregularities • Micro-roughness characteristics • Impact on friction and wear	L & Ds	PPT, PDF	20 min
(c)	• Surface Roughness Parameters • Ra and Rz definitions • Relation to Geometrical Surface • Effective Surface concept • Practical implications in maintenance	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between a real surface and a geometrical surface.
(b) How do long wavelength irregularities differ from short wavelength irregularities?
(c) Why is the effective surface important in engineering metrology?

12. Home Assignment and Follow Up	Prepare a brief note on how surface roughness affects the sealing efficiency of aircraft hydraulic components, referencing LP 02 concepts.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Standard maintenance procedure, inspection, standard maintenance and safety precaution	Refer to the master syllabus reference list	LP 11

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Flight Lieutenant, P No. 12345	Senior Instructor, Major, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min